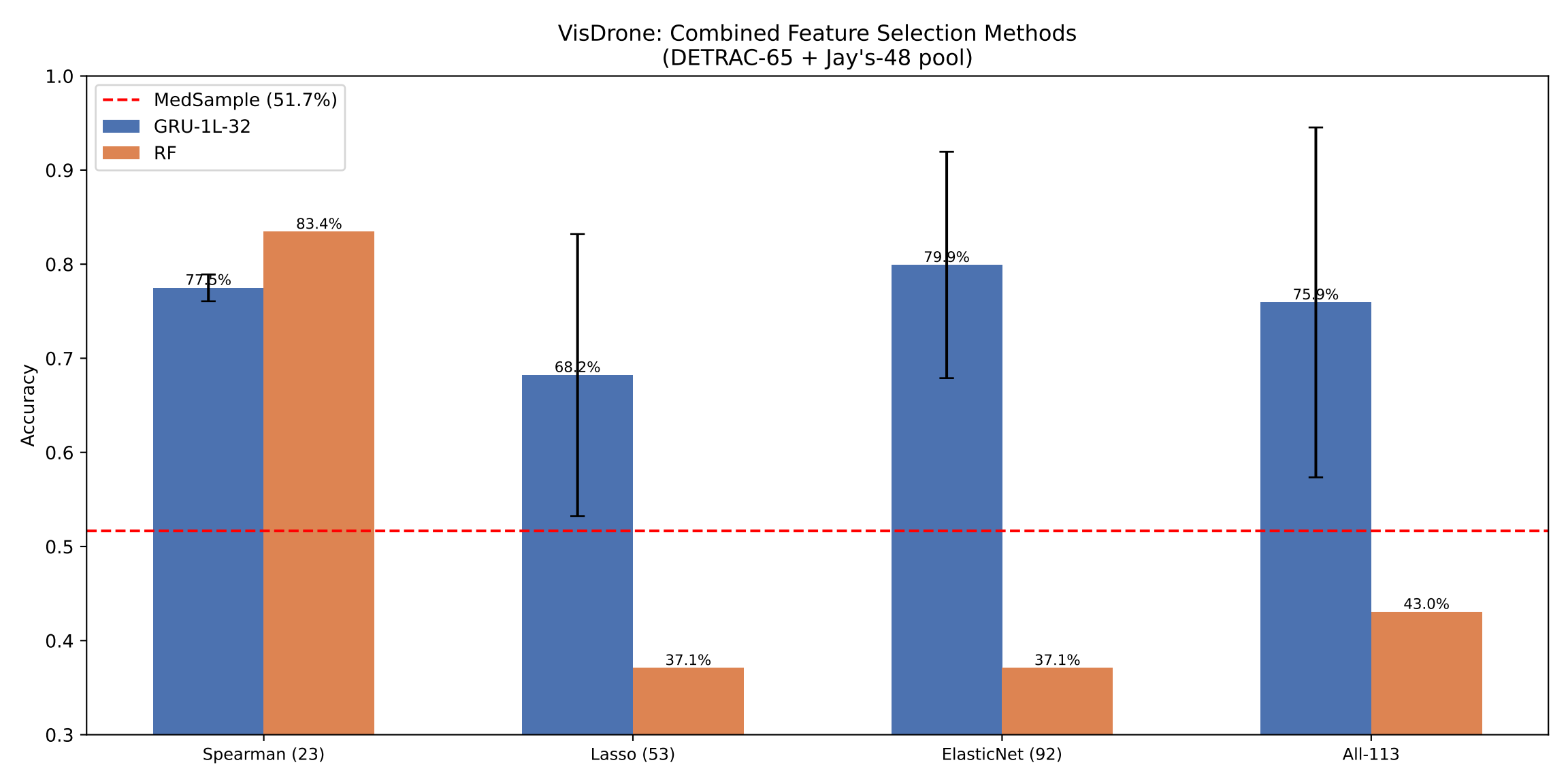
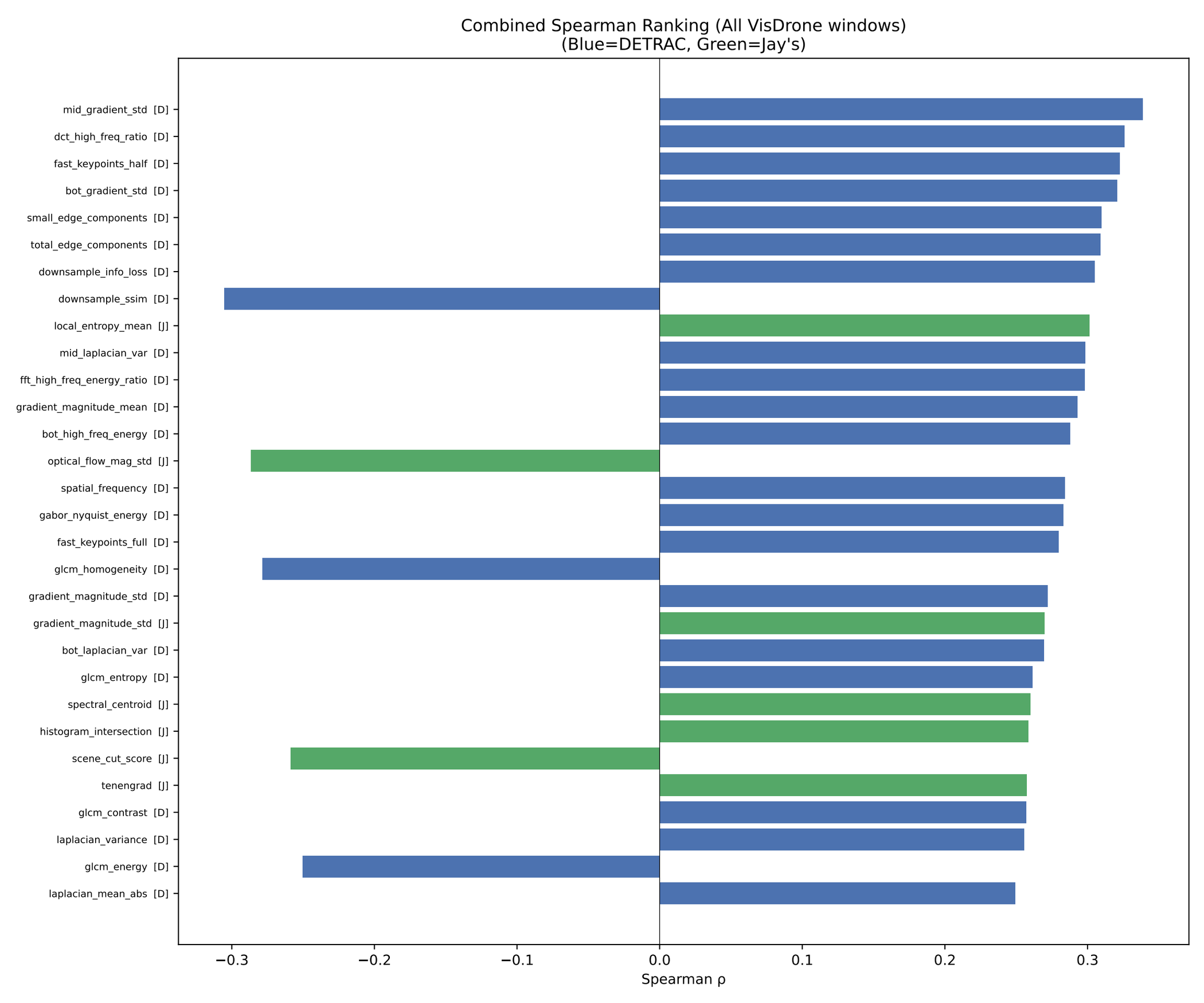
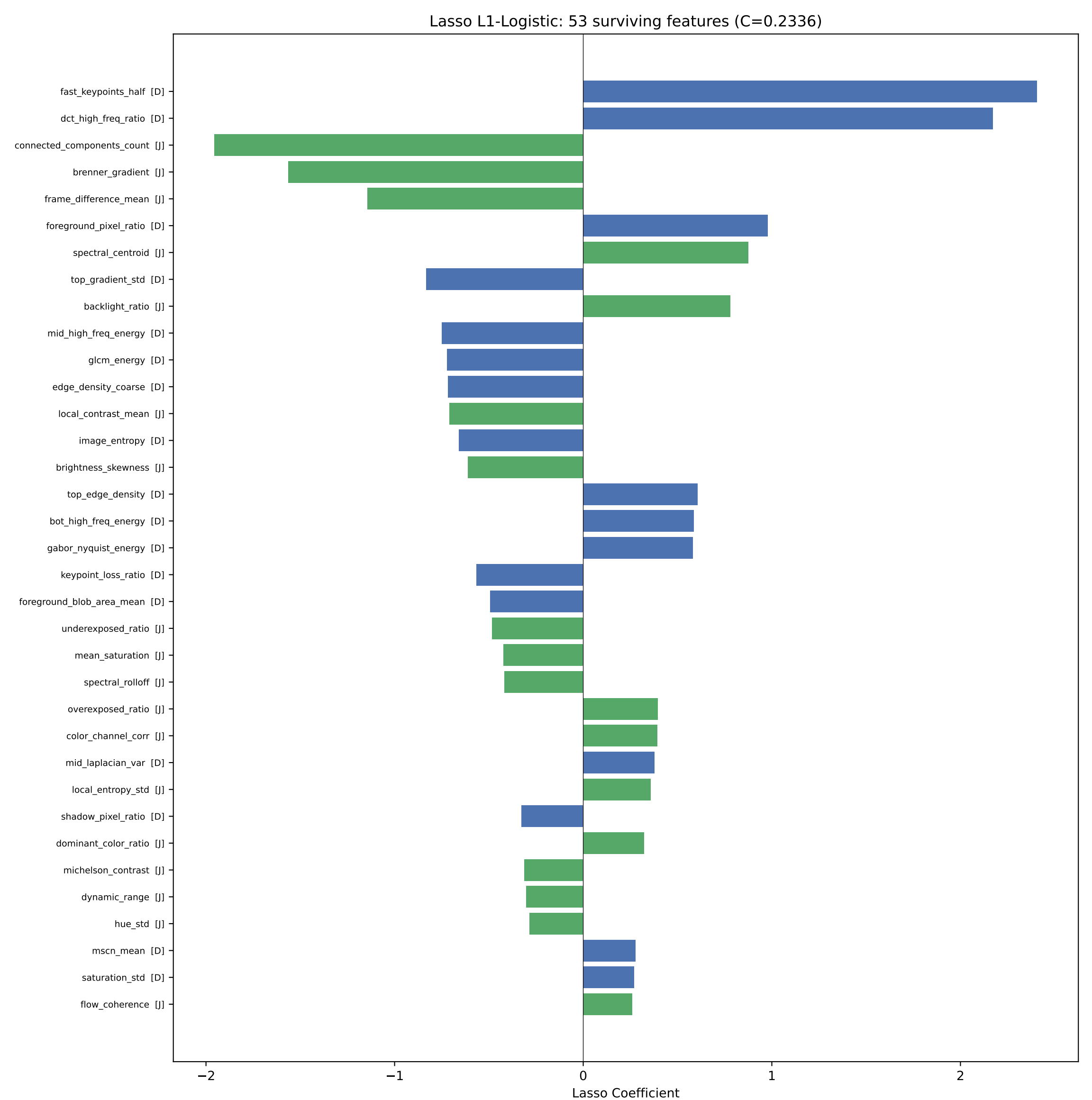

VisDrone: Combined Feature Selection Methods
(DETRAC-65 + Jay's-48 pool)
MedSample (51.7%)
GRU-1L-32
RF
Accuracy
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
77.5%
83.4%
68.2%
37.1%
79.9%
37.1%
75.9%
43.0%
Spearman (23)
Lasso (53)
ElasticNet (92)
All-113


Combined Spearman Ranking (All VisDrone windows)
(Blue=DETRAC, Green=Jay's)
mid_gradient_std [D]
dct_high_freq_ratio [D]
fast_keypoints_half [D]
bot_gradient_std [D]
small_edge_components [D]
total_edge_components [D]
downsample_info_loss [D]
downsample_ssim [D]
local_entropy_mean [J]
mid_laplacian_var [D]
fft_high_freq_energy_ratio [D]
gradient_magnitude_mean [D]
bot_high_freq_energy [D]
optical_flow_mag_std [J]
spatial_frequency [D]
gabor_nyquist_energy [D]
fast_keypoints_full [D]
glcm_homogeneity [D]
gradient_magnitude_std [D]
gradient_magnitude_std [J]
bot_laplacian_var [D]
glcm_entropy [D]
spectral_centroid [J]
histogram_intersection [J]
scene_cut_score [J]
tenengrad [J]
glcm_contrast [D]
laplacian_variance [D]
glcm_energy [D]
laplacian_mean_abs [D]
−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3
Spearman ρ

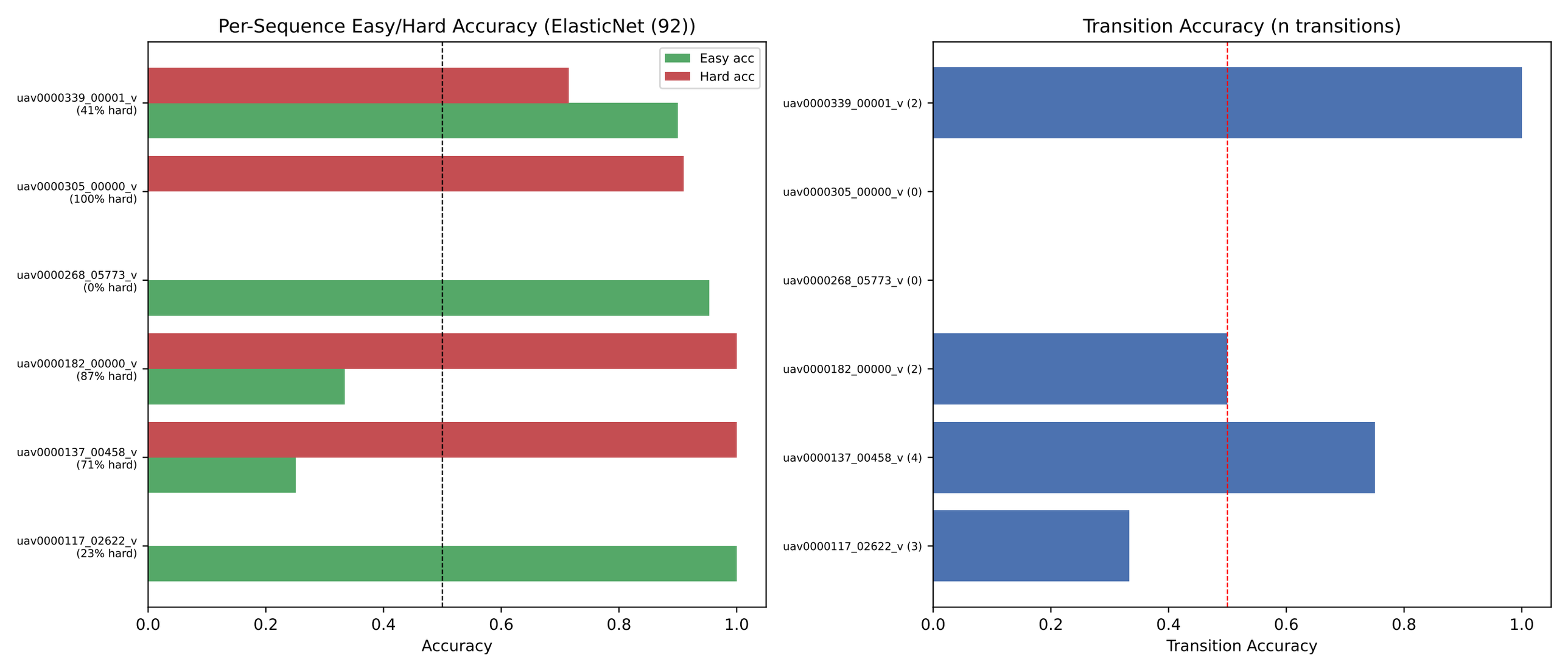

Lasso L1-Logistic: 53 surviving features (C=0.2336)
fast_keypoints_half [D]
dct_high_freq_ratio [D]
connected_components_count [J]
brenner_gradient [J]
frame_difference_mean [J]
foreground_pixel_ratio [D]
spectral_centroid [J]
top_gradient_std [D]
backlight_ratio [J]
mid_high_freq_energy [D]
glcm_energy [D]
edge_density_coarse [D]
local_contrast_mean [J]
image_entropy [D]
brightness_skewness [J]
top_edge_density [D]
bot_high_freq_energy [D]
gabor_nyquist_energy [D]
keypoint_loss_ratio [D]
foreground_blob_area_mean [D]
underexposed_ratio [J]
mean_saturation [J]
spectral_rolloff [J]
overexposed_ratio [J]
color_channel_corr [J]
mid_laplacian_var [D]
local_entropy_std [J]
shadow_pixel_ratio [D]
dominant_color_ratio [J]
michelson_contrast [J]
dynamic_range [J]
hue_std [J]
mscn_mean [D]
saturation_std [D]
flow_coherence [J]
−2
−1
0
1
2
Lasso Coefficient

Elastic Net: 92 surviving features (C=0.0336, l1_ratio=0.10)

connected_components_count [J]
fast_keypoints_half [D]
dct_high_freq_ratio [D]
backlight_ratio [J]
bot_high_freq_energy [D]
brenner_gradient [J]
top_edge_density [D]
keypoint_loss_ratio [D]
local_contrast_mean [J]
fast_keypoints_full [D]
brightness_skewness [J]
foreground_pixel_ratio [D]
mid_high_freq_energy [D]
frame_difference_mean [J]
spectral_centroid [J]
top_gradient_std [D]
mid_gradient_std [D]
foreground_blob_area_mean [D]
color_channel_corr [J]
SSIM_prev_frame [J]
mscn_mean [D]
fft_high_freq_energy_ratio [D]
underexposed_ratio [J]
foreground_blob_count [D]
mscn_std [D]
small_edge_components [D]
total_edge_components [D]
optical_flow_mag_std [J]
overexposed_ratio [J]
michelson_contrast [J]
edge_density_coarse [D]
flow_coherence [J]
downsample_ssim [D]
downsample_info_loss [D]
glcm_energy [D]

−0.8 −0.6 −0.4 −0.2 0.0 0.2 0.4 0.6

Elastic Net Coefficient


Per-Sequence Easy/Hard Accuracy (ElasticNet (92))
Easy acc
Hard acc
uav0000339_00001_v
(41% hard)
uav0000305_00000_v
(100% hard)
uav0000268_05773_v
(0% hard)
uav0000182_00000_v
(87% hard)
uav0000137_00458_v
(71% hard)
uav0000117_02622_v
(23% hard)
0.0
0.2
0.4
0.6
0.8
1.0
Accuracy
Transition Accuracy (n transitions)
uav0000339_00001_v (2)
uav0000305_00000_v (0)
uav0000268_05773_v (0)
uav0000182_00000_v (2)
uav0000137_00458_v (4)
uav0000117_02622_v (3)
0.0
0.2
0.4
0.6
0.8
1.0
Transition Accuracy